

Graph Theory Lecture Notes

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1 Lecture 1 (October 2nd 2026)

Definition 1.1. A *Graph* G is (V, Γ) where V is the set of vertices and Γ is the set of edges where $\Gamma \subseteq P_2(V)$.

There is a more rigorous definition for a graph but it is a little bit too general, obtuse, and keeps us from doing things intuitively.

Definition 1.2. A *Graph* G is (V_G, E_G, ϵ_G) where ϵ_G attaches edges to the vertex pair so we have $\epsilon_G : E_G \rightarrow \{\{u, v\} : u, v \in V_G\}$.

So to get some examples flowing so you understand, we have $\epsilon_G(e) = \{u, u\}$ then we know that e is a loop. Additionally if we have e_1, e_2 such that $\epsilon_G(e_1) = \epsilon_G(e_2)$ then we know that e_1, e_2 are double edges.

Definition 1.3. A graph is called *simple* if it has no loops or double edges.

I think we will be only looking at simple graphs. We are now going to introduce homomorphisms of graphs. There is some abuse of notation that I think will come up a lot here (or at least this seemed to be what the professor implied). We will typically write a homomorphism between $G = (V_G, E_G, \dots)$ and $H = (V_H, E_H, \dots)$ as $\phi : G \rightarrow H$ but in reality we will have $\phi_V : V_G \rightarrow V_H$ such that we respect edges. This means for all $u, v \in V_G$ we have $uE_Gv \implies \phi(u)E_H\phi(v)$. Hence,

Definition 1.4. A *graph homomorphism*, $\phi : G \rightarrow H$, is a map between the vertices of the graphs (V_G, V_H) such that for all $u, v \in V_G$ we have $uE_Gv \implies \phi(u)E_H\phi(v)$.

Exercise 1.5. Check that for two graph homomorphisms their composition is also a graph homomorphism.

Proof. Placeholder □

Definition 1.6. A graph homomorphism $\phi : G \rightarrow H$ is called an *isomorphism* if ϕ_V is bijective/invertible and the inverse, ϕ_V^{-1} is also a homomorphism.

Example 1.7. Here is a **non-example**: the map from two vertices with no edges to a map of two vertices with an edge is a homomorphism and a bijection but the inverse is not.

We now have some example graphs we need to describe for notations sake, and we explain what the complement of a graph looks like and is notated as.

Definition 1.8. The complement of a graph G , written as $\overline{G}$ is defined as $(V_G, \dots)$ but we have $u\overline{E}_Gv \iff$ we do not have uE_Gv

We now have some examples:

1. $K_1 = \overline{K_1}$ is the trivial graph of just one point.
2. K_n is called a clique it has the property that $\forall u, v \in G$ we have uE_Gv this is also called the complete graph on n points.
3. $\overline{K_n}$ is called an independent graph on n vertices. and has the property that for all $u, v \in G$ we do not have uE_Gv this is also called the null graph on n points.
4. We also have the path graph π which looks like $v_1 \rightarrow v_2 \rightarrow v_3 \cdots \rightarrow v_n$ so n vertices and $n - 1$ edges. We write, $V = [n] := \{1, \dots, n\}$ with edges $(i)E_\pi(i + 1)$ for $i \in \{1, \dots, n - 1\}$.
5. We also have a C_n cycle which is the same as a path but with an edge between v_n and v_1 so you can define it using modular arithmetic on the last path graph we wrote.

We also now define bi-partite graphs, which are basically graphs with two separate sets of vertices with only connections between these two collections and not amongst the collections themselves. We also have n -partite graphs that have n collections instead of just 2. Extend the definition below similarly for n -partite graphs.

Definition 1.9. A *bipartite graph*, G , is a graph with subsets of vertices $U_1, U_2 \subseteq V_G, U_1 \neq \emptyset \neq U_2$ where we have $U_1 \cap U_2 = \emptyset$ and $U_1 \cup U_2 = V_G$ and for any $u, v \in U_1 \implies$ we do not have uEv .